

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

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Lab Objectives

- Identify code duplication and inefficiencies
- Refactor code using AI assistance
- Improve readability and maintainability
- Apply Python best practices

Task 1: Removing Code Duplication

Legacy Code

```
print("Area:", 5 * 10)
print("Perimeter:", 2 * (5 + 10))
print("Area:", 7 * 12)
print("Perimeter:", 2 * (7 + 12))
```

Refactored Code

```
def rectangle_metrics(length, width):
    area = length * width
    perimeter = 2 * (length + width)
```

```
        return area, perimeter

print(rectangle_metrics(5, 10))
print(rectangle_metrics(7, 12))
```

Task 2: Optimizing Loops

Legacy Code

```
names = ["Alice", "Bob"]
search = ["Bob", "Eve"]

for s in search:
    found = False
    for n in names:
        if s == n:
            found = True
    if found:
        print("Found")
```

Refactored Code

```
names = {"Alice", "Bob"}
search = ["Bob", "Eve"]

for s in search:
    if s in names:
        print("Found")
```

Task 3: Extracting Functions

```
def calculate_total(price):
    tax = price * 0.18
    return price + tax

print(calculate_total(250))
print(calculate_total(500))
```

Task 4: Using Constants

```
PI = 3.14159
```

```
RADIUS = 7

area = PI * (RADIUS ** 2)
circumference = 2 * PI * RADIUS

print(area)
print(circumference)
```

Task 5: Improving Naming

```
base = 10
height = 20
area_of_triangle = base * height / 2

print(area_of_triangle)
```

Conclusion

Refactoring improves code readability, reduces duplication, and enhances maintainability.
AI tools assist in identifying improvements efficiently.